DB-Lab Grand Task

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Q1-6:

```
create table customers(
    customer_id int,
    first_name varchar(50) not null,
    last_name varchar(50) not null,
    email varchar(100),
    phone varchar(15) not null,
    city varchar(50),
    registration_date date,

    primary key(customer_id)
    );

create table employees(
    employee_id int,
    manager_id int,
    first_name varchar(50) not null,
    last_name varchar(50) not null,
    job_title varchar(50) not null,
    salary decimal(10, 2) not null,
    hire_date date,

    primary key(employee_id)
    );

create table suppliers(
    supplier_id int,
    supplier_name varchar(100) not null,
    contact_email varchar(100),
    city varchar(50),

    primary key(supplier_id)
    );
```

```sql
create table products(
    product_id int,
    product_name varchar(100) not null,
    category varchar(50),
    price decimal(10, 2),
    stock_quantity int,
    supplier_id int,

    primary key(product_id),
    foreign key(supplier_id) references suppliers(supplier_id)
    );

create table orders(
    order_id int,
    customer_id int,
    employee_id int,
    order_date date,
    status varchar(30),

    primary key(order_id),
    foreign key(customer_id) references customers(customer_id),
    foreign key(employee_id) references employees(employee_id)
    );

create table order_details(
    order_detail_id int,
    order_id int,
    product_id int,
    quantity int,
    unit_price decimal(10,2),

    primary key(order_detail_id),
    foreign key(order_id) references orders(order_id),
    foreign key(product_id) references products(product_id)
    );
```

```sql
alter table products
modify product_name varchar(100) not null;

alter table customers
modify email varchar(100) unique;

alter table employees
modify salary decimal(10, 2) check(salary>10000);
```

```sql
alter table products
modify price decimal(12, 2);

alter table suppliers
drop city;
```

Q7-12:

```sql
insert into customers(customer_id, first_name, last_name, email, phone, city, registration_date) values
    (1,'Ali','Khan','ali.khan@gmail.com','03001234567','Karachi','2023-01-10'),
    (2,'Sara','Ahmed','sara.ahmed@gmail.com','03111234567','Lahore','2022-06-15'),
    (3,'Usman','Raza','usman.raza@gmail.com','03221234567','Islamabad','2024-02-20'),
    (4,'Ayesha','Malik','ayesha.malik@gmail.com','03331234567','Karachi','2023-08-12'),
    (5,'Hassan','Iqbal','hassan.iqbal@gmail.com','03451234567','Lahore','2021-11-01');

insert into employees(employee_id, first_name, last_name, job_title, salary, hire_date, manager_id) values
    (1,'Kamran','Ali','General Manager',120000,'2020-02-01',NULL),
    (2,'Sadia','Khan','Sales Manager',80000,'2021-05-10',1),
    (3,'Imran','Shah','Sales Executive',50000,'2022-03-15',2),
    (4,'Nida','Rauf','Sales Executive',48000,'2023-01-20',2),
    (5,'Tariq','Mehmood','Accountant',60000,'2021-09-12',1);

insert into suppliers(supplier_id, supplier_name, contact_email, city) values
    (1,'TechSource Ltd','contact@techsource.com','Karachi'),
    (2,'Global Electronics','info@globalelec.com','Lahore'),
    (3,'HomeNeeds Supplier','sales@homeneeds.com','Islamabad'),
    (4,'Fashion Hub Distributors','support@fashionhub.com','Karachi'),
    (5,'OfficeWorld Traders','office@officeworld.com','Lahore');
```

```sql
insert into products(product_id, product_name, category, price, stock_quantity, supplier_id) values
    (1,'Laptop Pro 15','Electronics',150000,20,1),
    (2,'Wireless Mouse','Electronics',2500,100,2),
    (3,'Office Chair','Furniture',18000,50,5),
    (4,'LED TV 42 Inch','Electronics',85000,15,2),
    (5,'Men T-Shirt','Clothing',2000,200,4),
    (6,'Refrigerator 300L','Appliances',95000,10,3),
    (7,'Smartphone X','Electronics',120000,30,1),
    (8,'Wooden Desk','Furniture',25000,25,5);

insert into orders(order_id, customer_id, employee_id, order_date, status) values
    (1,1,3,'2024-01-15','Completed'),
    (2,2,4,'2024-02-10','Completed'),
    (3,3,3,'2024-03-05','Pending'),
    (4,4,4,'2023-12-20','Completed'),
    (5,5,3,'2024-04-01','Shipped');

insert into order_details(order_detail_id, order_id, product_id, quantity, unit_price) values
    (1,1,1,1,150000),
    (2,1,2,2,2500),
    (3,2,4,1,85000),
    (4,2,5,3,2000),
    (5,3,7,1,120000),
    (6,4,3,2,18000),
    (7,5,6,1,95000),
    (8,5,2,1,2500),
    (9,6,9,1,110000),
    (10,7,8,1,25000);
```

Update / Delete:

```
update employees
set salary=salary*1.1
where hire_date<2022
```

```
update orders
set status='Completed'
where order_date<2024
```

Show query box

9 rows affected. (Query took 0.0004 seconds.)

```
UPDATE product_table AS p JOIN order_detail_table AS od ON p.product_id = od.product_id SET p.stock_quantity = p.stock_quantity - od.quantity;
```

[Edit inline] [Edit] [Create PHP code]

Show query box

0 rows affected. (Query took 0.0005 seconds.)

```
UPDATE order_table AS o SET o.status = 'Completed' WHERE o.order_date < '2024-01-01';
```

[Edit inline] [Edit] [Create PHP code]

Aggregate Functions:

Showing rows 0 - 0 (1 total, Query took 0.0001 seconds.)

```
select count(customer_id) from customer_table;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Showing rows 0 - 0 (1 total, Query took 0.0002 seconds.)

```
select avg(salary) from employee_table as average_salary;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Showing rows 0 - 0 (1 total, Query took 0.0001 seconds.)

```
select max(price) from products_table as hgihest_price;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Showing rows 0 - 6 (7 total, Query took 0.0002 seconds.)

```
select sum(quantity * unit_price) from order_details_table group by order_id;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Showing rows 0 - 3 (4 total, Query took 0.0002 seconds.)

```
select category,sum(stock_quantity) from products_table GROUP by category;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Your SQL query has been executed successfully.

```
SELECT category, COUNT(product_id) AS product_count FROM product_table GROUP BY category HAVING COUNT(product_id) > 3;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Showing rows 0 - 1 (2 total, Query took 0.0025 seconds.)

```
SELECT employee_id, COUNT(order_id) AS total_orders FROM order_table GROUP BY employee_id;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Showing rows 0 - 0 (1 total, Query took 0.0005 seconds.)

```
SELECT first_name, last_name, salary FROM employee_table WHERE salary = (SELECT MAX(salary) FROM employee_table);
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Character Functions:

Showing rows 0 - 6 (7 total, Query took 0.0004 seconds.)

```
SELECT UPPER(CONCAT(c.`first_name`, ' ', c.`last_name`)) AS full_name_upper FROM customer_table AS c;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Showing rows 0 - 8 (9 total, Query took 0.0004 seconds.)

```
SELECT SUBSTRING(p.product_name, 1, 3) AS short_name FROM product_table AS p;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Showing rows 0 - 5 (6 total, Query took 0.0005 seconds.)

```
SELECT CONCAT(e.first_name, ' ', e.`last_name`) AS employee_full_name FROM employee_table AS e;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Showing rows 0 - 4 (5 total, Query took 0.0004 seconds.)

```
SELECT s.supplier_name, LENGTH(s.supplier_name) AS name_length FROM supplier_table AS s;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

✔ Showing rows 0 - 6 (7 total, Query took 0.0004 seconds.)

```sql
SELECT LOWER(c.email) AS lowercase_email FROM customer_table AS c;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

✔ Showing rows 0 - 2 (3 total, Query took 0.0005 seconds.)

```sql
SELECT * FROM customer_table AS c WHERE c.`first_name` LIKE 'A%';
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Date Functions:

✔ Showing rows 0 - 6 (7 total, Query took 0.0004 seconds.)

```sql
SELECT c.`first_name`, c.`last_name`, DATEDIFF(CURDATE(), c.registration_date) AS days_since_registration FROM customer_table AS c;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

✔ Showing rows 0 - 1 (2 total, Query took 0.0004 seconds.)

```sql
SELECT * FROM employee_table AS e WHERE YEAR(e.`hire_date`) = 2023;
```

✔ Showing rows 0 - 4 (5 total, Query took 0.0004 seconds.)

```sql
SELECT e.first_name, e.`last_name`, e.`hire_date` FROM employee_table AS e WHERE TIMESTAMPDIFF(YEAR, e.`hire_date`, CURDATE()) > 2;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

✔ Showing rows 0 - 0 (1 total, Query took 0.0002 seconds.)

```sql
SELECT CURDATE() AS current_system_date;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Operators:

✔ Showing rows 0 - 1 (2 total, Query took 0.0005 seconds.)

```sql
SELECT * FROM product_table AS p WHERE p.price BETWEEN 1000 AND 5000;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

✔ Showing rows 0 - 0 (1 total, Query took 0.0004 seconds.)

```sql
SELECT * FROM employee_table AS e WHERE e.salary IN (30000, 40000, 50000);
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Showing rows 0 - 2 (3 total, Query took 0.0005 seconds.)

```sql
SELECT * FROM customer_table AS c WHERE c.city NOT IN ('Karachi', 'Lahore');
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Showing rows 0 - 3 (4 total, Query took 0.0006 seconds.)

```sql
SELECT * FROM product_table AS p WHERE p.category LIKE '%Electronics%';
```

Showing rows 0 - 5 (6 total, Query took 0.0005 seconds.)

```sql
SELECT * FROM order_table AS o WHERE o.`order_date` BETWEEN '2024-01-01' AND '2024-12-31';
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Showing rows 0 - 10 (11 total, Query took 0.0007 seconds.)

```sql
SELECT c.`first_name`, c.`last_name`, od.* FROM customer_table AS c JOIN order_table AS o ON c.customer_id = o.customer_id JOIN order_detail_table AS od ON o.`order_id` = od.`order_id`;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

```sql
SELECT p.product_name, s.supplier_name
FROM product_table AS p
JOIN supplier_table AS s ON p.supplier_id = s.supplier_id;
```

Showing rows 0 - 6 (7 total, Query took 0.0004 seconds.)

```sql
SELECT o.`order_id`, e.first_name, e.`last_name` FROM order_table AS o JOIN employee_table AS e ON o.employee_id = e.employee_id;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Showing rows 0 - 7 (8 total, Query took 0.0006 seconds.)

```sql
SELECT c.`first_name`, c.`last_name`, o.`order_id` FROM customer_table AS c LEFT JOIN order_table AS o ON c.customer_id = o.customer_id;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Showing rows 0 - 5 (6 total, Query took 0.0005 seconds.)

```sql
SELECT e1.first_name AS employee_name, e2.first_name AS manager_name FROM employee_table AS e1 LEFT JOIN employee_table AS e2 ON e1.employee_i
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Showing rows 0 - 1 (2 total, Query took 0.0007 seconds.)

```sql
SELECT e.first_name, e.`last_name`, SUM(od.quantity * od.unit_price) AS total_sales FROM employee_table AS e JOIN order_table AS o ON e.employ
o.`order_id` = od.`order_id` GROUP BY e.employee_id;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Showing rows 0 - 2 (3 total, Query took 0.0005 seconds.)

```sql
SELECT p.product_name, SUM(od.quantity) AS total_sold FROM product_table AS p JOIN order_detail_table AS od ON p.product_id = od.product_id GROUP BY p.product_id ORDER BY total_sold DESC LIMIT 3;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]